

Structural-functional brain network modelling using non-linear, stochastic Gaussian and diffusion model to compare information cost assumption between ideal and Empirical network

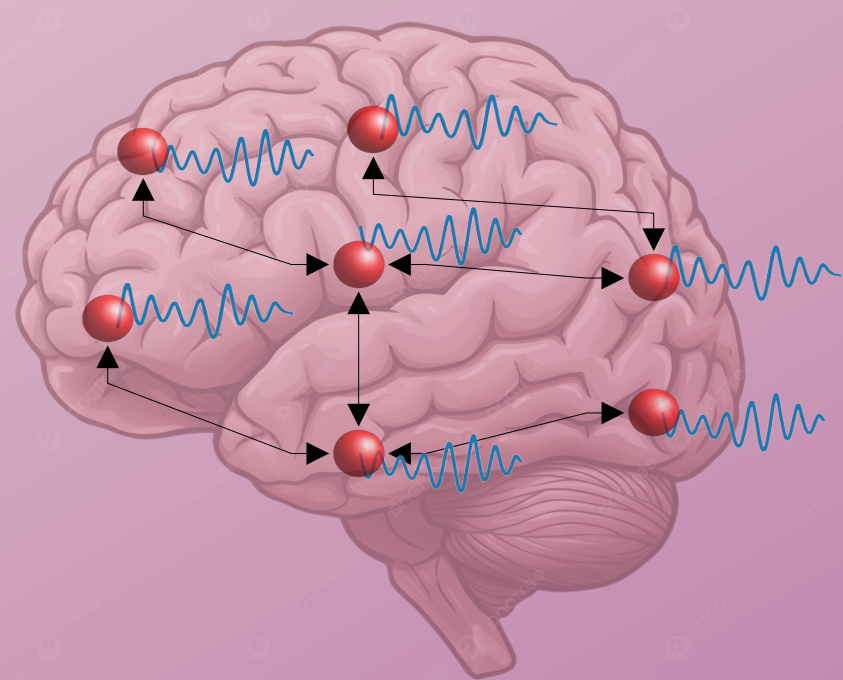


Figure 1: Structural (black arrows) – functional (blue activity) illustration of brain clusters

Introduction

Brain network modelling is the area of research in which the macroscopic behaviour of the brain is modelled. The goal is to understand behaviours of multiple neurons or neuron clusters and their interaction.

In this project the signalling behavior of 246 brain clusters (nodes) are modelled using 3 different methods for the structural-functional interaction. This aims to describe the relation between the physical connections (white matter) and brain activity during resting state.

For various brain network models three important metrics are delay, energy and information cost. These are approximate evaluations of how well the model represents individual neuronal behaviour.

The question we aim to answer is whether the assumption that brain networks are primarily built using shortest path assumption. A assumption which is of high information cost as individual neuronal behaviour is highly decentralised and does not have access to global characteristics of the brain.

100 individuals lay down in a MRI scanner and told to not to think of anything, collecting their anatomical connectome mapping (structural DTI) and brain region activity (fMRI), which is represented by figure 1. This is also represented in figure 2 as a connectome map and figure 3 as the empirical brain activity time series.

Methodology

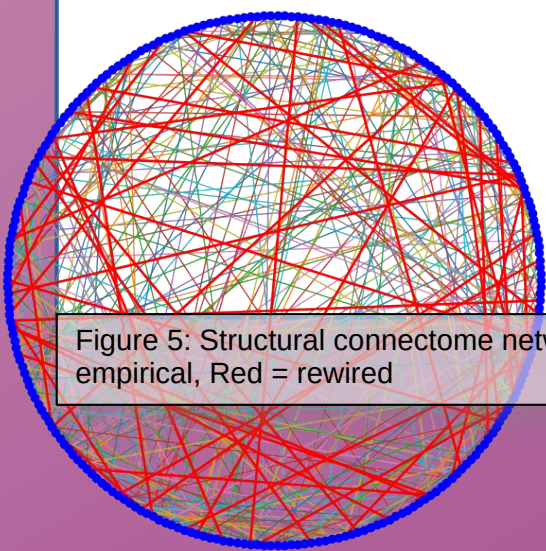


Figure 5: Structural connectome network graph Black = empirical, Red = rewired

Activity – time plot for:
Empirical data – one node
Models 3x – one node

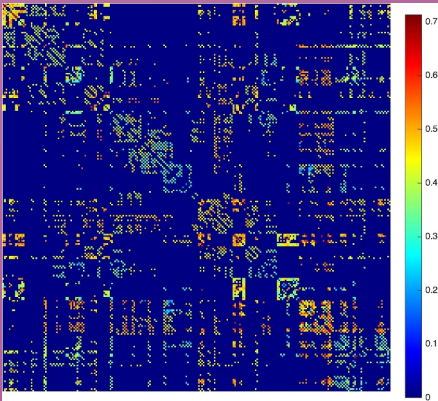


Figure 3: Activity- time plot for empirical and simulated signals

Figure 2: connectome example

Figure showing Pearson coefficient of the correlation between each model prediction and the true functional connectivity.
matrix rsFC (3 subplots looking like figure)

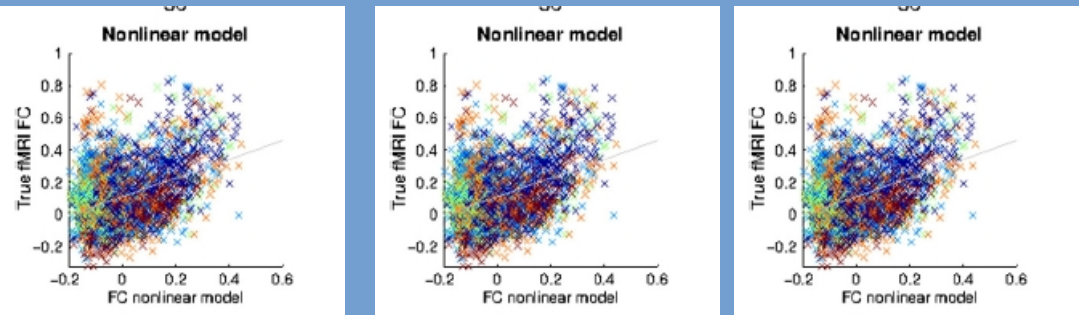


Figure 4: Simulated and empirical correlation

Figure Showing the difference due to reconnection of connectome

Figure 6: Results due to reconnection of connectome

Results and discussion